

IGATE's Live Remote Assistance app enabled by wearable smart glasses

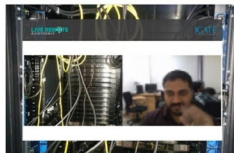
During the field visits of a service engineer, referring to a product manual or using mobile assistance to service particular equipment might not work due to the inexperience of the engineer. Experts in a remote location, who do not have a direct view of the equipment or device being serviced, will not be helpful too. To overcome these difficulties, IGATE has developed a 'Live Remote Assistance' wearable glass application, that provides the assistance of a remotely located expert to field technicians. This application can be used in manufacturing, logistics, utilities and other industries.

IGATE has used Epson Moverio BT-200 wearable smart glasses, and the application makes use of its camera through which a live feed is sent over the Internet to the remote expert, who then guides the user by voice and video. The expert can provide better assistance to the field technicians with a live view provided by the wearable glass camera.

Video and voice collaboration with experts in remote locations using Live Remote Assistance would result in faster repairs and save the expense of flying in an expert to the site for help. These collaborations can be recorded and stored as reference material for future jobs, as well as video evidence for use in future in-depth investigations.



Live Remote Assistance for network troubleshooting



Live Remote Assistance wearable glass view

Leading market players offering WebRTC as a service

As WebRTC has gained wider popularity, the opportunity to offer it as a service has risen. Featured here are some of the more well-known options.

PubNub

PubNub offers a data stream network that can be used as a scalable signalling server for WebRTC applications. PubNub features Presence and storage/playback, which can be used to enhance applications.

The WebRTC protocol does not provide storage capabilities, and as a result, there are no records of what messages have been sent. Specifically with text chat, users expect a history of previous chat conversations. PubNub's storage/playback feature allows users to see a history of past conversations over a desired period of time.

Bistri

Bistri was the very first service to provide a production platform for the implementation of WebRTC applications. Bistri harnesses WebRTC to provide users with a personalised link that, when clicked, instantly initiates a video call, like an online phone number. There is no software, plugins or accounts required; the call is carried out completely through the browser. A single snippet of JavaScript is all that is needed to attach a button to a website or blog, offering visitors a simple click-to-video-call experience.

TokBox

TokBox is a PaaS (Platform as a Service) company that provides hosted infrastructure, APIs and the tools required to deliver enterprise-grade WebRTC capabilities. It does so primarily through its proprietary OpenTok video platform for commercial application.

The OpenTok platform provides APIs, a global cloud infrastructure and pre-configured solutions to enterprises, entrepreneurs and developers. Originally launched as an Adobe Flash based platform in 2010, OpenTok became the first real-time communications platform to incorporate support for WebRTC in 2012. In mid-2013, TokBox launched Mantis, a cloud based infrastructure enabling multi-party conversations on the OpenTok platform.

Developers use the OpenTok platform to deploy WebRTC applications with 10x fewer lines of code than required through WebRTC off-the-shelf. OpenTok also supports multi-party calling, calls between browsers and iOS or Android devices, call archiving and playback, secure enterprise firewalls traversal and more.

Challenges in adopting WebRTC

Implementation of current WebRTC specifications has not yet been completed by different browser providers. Google Chrome, Firefox and Opera support WebRTC without any plugins. Chrome v29 or greater supports WebRTC on Android. Internet Explorer and Safari don't support WebRTC natively. Third party plugins can enable WebRTC features on these browsers to an extent.